Chapter 6, Problem 3PP

Problem

The current through a 100-μF capacitor is $i(t)$ = 50 sin 120πt mA. Calculate the voltage across it at t = 1 ms and t = 5 ms. Take $v(0)$ = 0.

Step-by-step solution

Step 1 of 3

Refer to equation 6.6 in the textbook.

The voltage across the capacitor is,

$$v=\frac{1}{C}\int_{t_0}^{t} idt+v(t_0)$$

Here, $v(t_0)$ is the voltage across the capacitor at time t_0 sec.

Substitute 0 for t_0.

$$v=\frac{1}{C}\int_{0}^{t} idt+v(0)$$

Substitute 0 for $v(0)$, 100×10^{-6} for *C*, and $50\times10^{-3}\sin(120\pi t)$ for *i*.

$$\begin{aligned}v&=\frac{1}{100\times10^{-6}}\int_{0}^{t}50\times10^{-3}\sin(120\pi t)dt+0\\&=\frac{50\times10^{-3}}{100\times10^{-6}}\int_{0}^{t}\sin(120\pi t)dt\\&=500\left[\frac{-\cos(120\pi t)}{120\pi}\right]_0^t\\&=\frac{500}{120\pi}\times\left[-\cos120\pi(t)-(-\cos120\pi(0))\right]\end{aligned}$$

Therefore,

$$v=1.3263\times[-\cos120\pi t+1] \quad \ldots\ldots (1)$$

Step 2 of 3

Recall equation (1).

$$v=1.3263\times[-\cos120\pi t+1]$$

Substitute 10^{-3} for *t*.

$$\begin{aligned}v&=1.3263\left[-\cos120\pi\times10^{-3}+1\right]\\&=1.3263(0.0702235)\\&=93.14\text{ mV}\end{aligned}$$

Therefore, the voltage across the capacitor at $t=1\text{ ms}$ is $\boxed{93.14\text{ mV}}$.

Step 3 of 3

Recall equation (1).

$$v = 1.3263 \times \left[-\cos 120\pi t + 1\right]$$

Substitute 5×10^{-3} for *t*.

$$\begin{aligned} v &= 1.3263\left[-\cos 120\pi \times 5 \times 10^{-3} + 1\right] \\ &= 1.3263(1.309016994) \\ &= 1.736\ \text{V} \end{aligned}$$

Therefore, the voltage across the capacitor at $t = 5\ \text{ms}$ is $\boxed{1.736\ \text{V}}$.